

VIETNAM NATIONAL UNIVERSITY-HO CHI MINH CITY

HO CHI MINH UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY

KNOWLEDGE ENGINEERING DEPARTMENT

Homework 2

Course: Introduction to Artificial Intelligence

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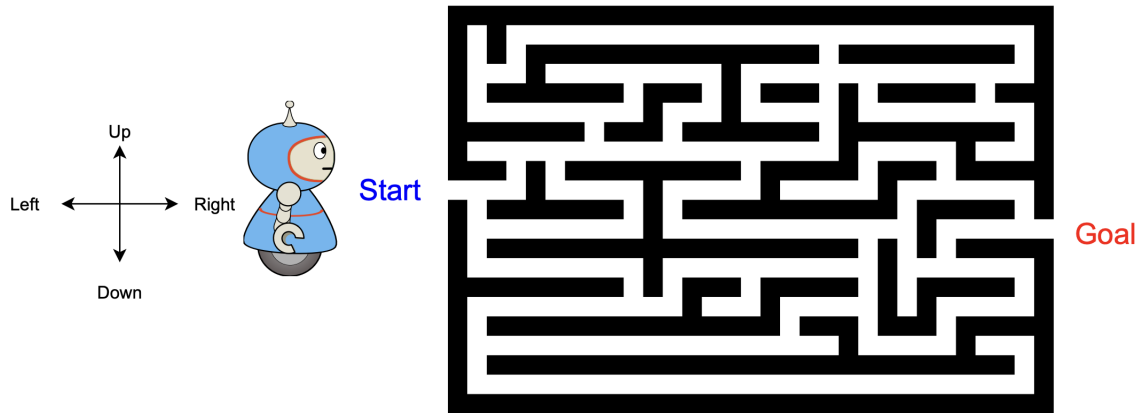


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1 Exercise

Given a maze as shown below. For each of the following **graph search** strategies, **paint** all of the space visited by the algorithms, as well as the returned solutions. (Note that the priority order of actions is RIGHT, DOWN, LEFT, UP)



- (a) Breadth-first search
- (b) Uniform-cost search
- (c) Depth-first search

2 Breadth-first search

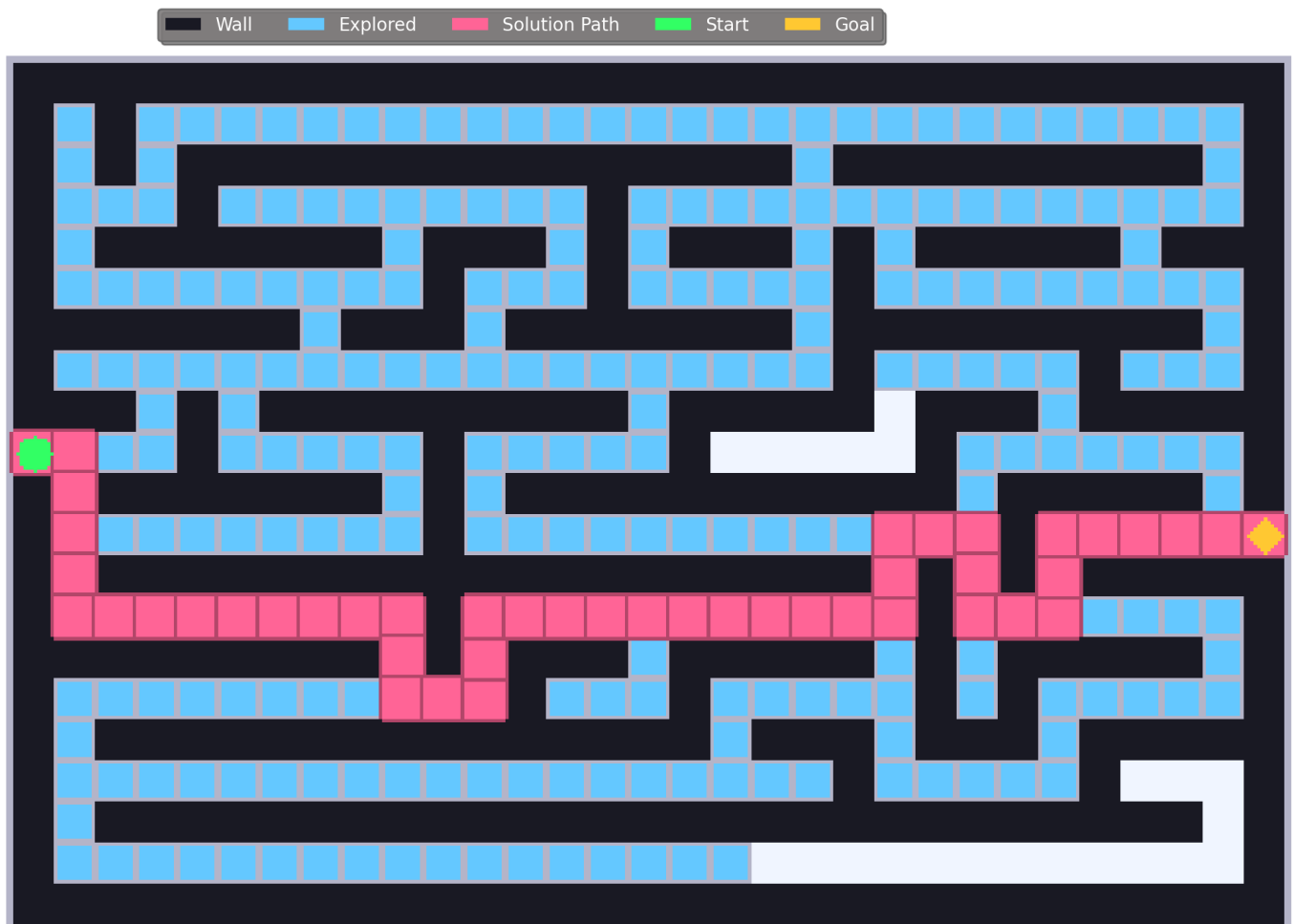
2.1 Algorithm

- Breadth-first search explores the maze layer by layer from the Start node. It uses a queue (FIFO) and expands all nodes at a given distance before moving to the next distance level. This guarantees that BFS will find the shortest path in terms of the number of steps. The search stops as soon as the Goal node is reached.

2.2 Implementation

BFS Pathfinding

Path: 45 steps • Explored: 289 nodes



3 Uniform-cost search

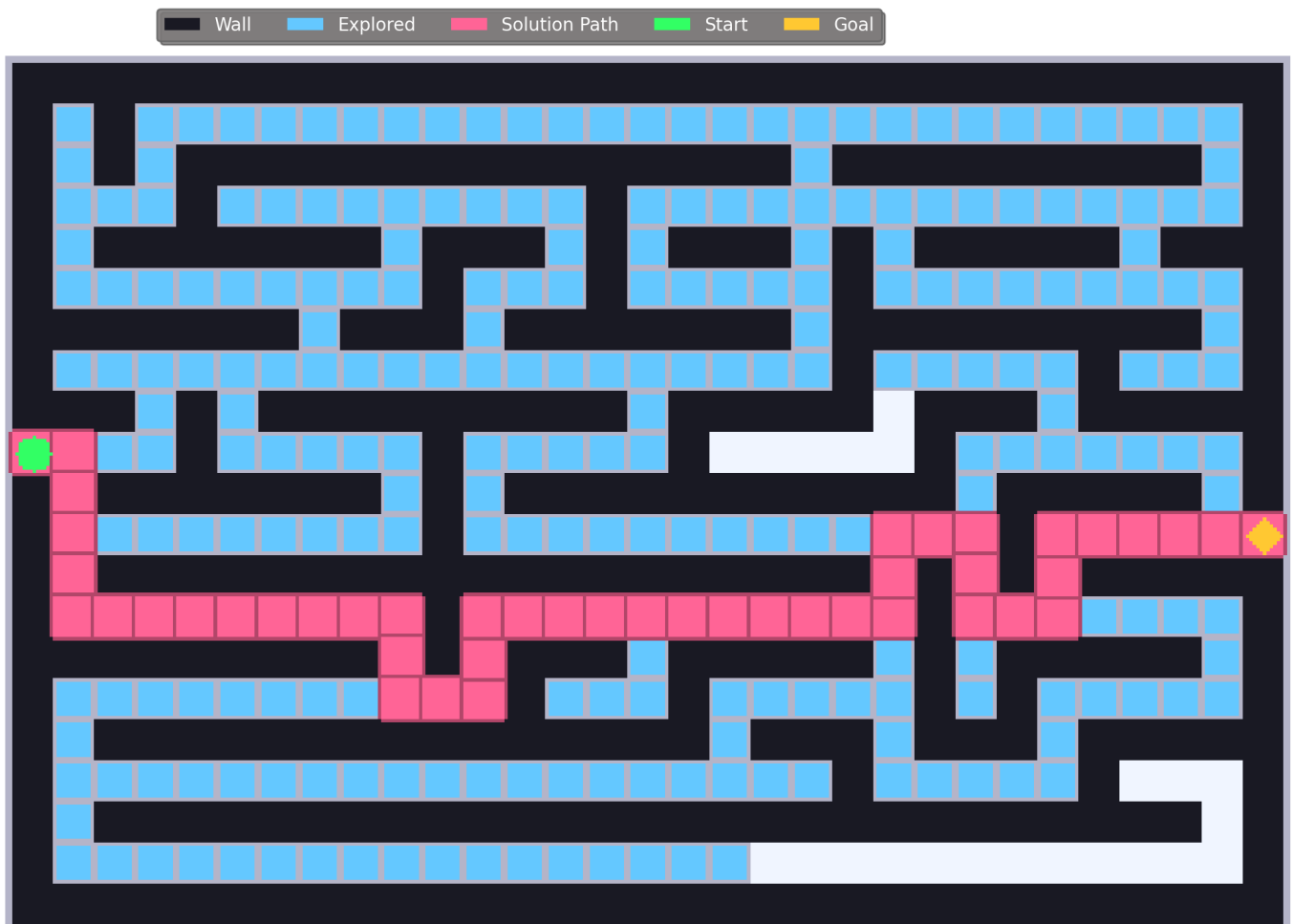
3.1 Algorithm

- Uniform-cost search expands nodes based on the lowest cumulative cost from the start. In this maze, every step (action) has a uniform cost of 1. When all step costs are equal, UCS behaves identically to Breadth-First Search. It will explore nodes in the same order as BFS, guaranteeing the discovery of the least-cost (and thus shortest) path.

3.2 Implementation

UCS Pathfinding

Path: 45 steps • Explored: 289 nodes



4 Depth-first search

4.1 Algorithm

- Depth-first search explores as deeply as possible along one path before backtracking. It uses a stack (LIFO). Following the specified priority (RIGHT, DOWN, LEFT, UP), the algorithm will always try to go RIGHT if possible. If not, it will try DOWN, then LEFT, then UP. This "dive-first" strategy will lead it down long corridors and into dead ends. The algorithm backtracks only when it can no longer move forward. The first path it finds to the Goal is returned, which is typically not the shortest path.

4.2 Implementation

DFS Pathfinding

Path: 49 steps • Explored: 286 nodes

